Name: ______________________

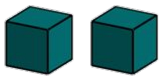
Worksheet

Problem solving with theoretical probability

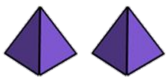
Task A: Using probability to inform decisions

1) Andeep is playing a game that involves choosing two dice, rolling them and summing their numbers.

Two of the dice are cubes containing numbers 1 to 6



Two of the dice are tetrahedrons containing numbers 1 to 4



a) Which combination of dice give the greatest probability of rolling a total of 8?

b) Which combination of dice give the greatest probability of rolling a total of 6?

c) Which combination of dice give the greatest probability of rolling a total of 7?

d) Write a summary that advises a player how to decide which pair of dice gives them the greatest probability of rolling the total they want.

Task B: Number of trials required to observe a difference

1) Lucas creates two versions of the same video game: Version A and Version B.
In Version A, the probability that a player finds a gem is 0.02
Players are 10% more likely to find a gem in Version B than in Version A.

a) Complete the table.

	Version A	**Version B**	**difference**
probability of finding gem	0.02		
expected gems per 100 turns			
expected gems per 1000 turns			

b) How many turns would it take to expect a difference of 1 gem between the two versions of the game?

2) Sofia creates a video game where players have a 0.5% probability of getting caught in a trap each turn.
She creates a new version of the game where players are now 20% more likely to get caught in a trap each turn, compared to the original version of the game.

a) What is the probability of getting caught in a trap in the new version of the game?

b) How many turns would it take to expect a difference of 1 extra trap between the two versions of the game?